Marwadi University	Marwadi University Faculty of Technology Department of Information and Communication Technology	
Subject: Digital Signal and Image Processing (01CT1513)	Aim : Simulate linear convolution and circular convolution on discrete time signals.	
Experiment -2	Date:25-07-2026	Enrolment No: 92400133079

Code 1:-

```python
import numpy as np
import matplotlib.pyplot as plt

def linear_convolution(signal1, signal2):
    linear_conv = np.convolve(signal1, signal2, mode='full')
    return linear_conv

def circular_convolution(signal1, signal2):
    if len(signal1) > len(signal2):
        fft_length = len(signal1)
    else:
        fft_length = len(signal2)

    fft_signal1 = np.fft.fft(signal1, fft_length)
    fft_signal2 = np.fft.fft(signal2, fft_length)

    circular_conv = np.fft.ifft(fft_signal1 * fft_signal2)

    return circular_conv

signal1 = np.array([1, 2, 3, 4, 5])
signal2 = np.array([2, 4, 6, 8, 10])

linear_conv = linear_convolution(signal1, signal2)
circular_conv = circular_convolution(signal1, signal2)

plt.figure(figsize=(10, 6))

plt.subplot(2, 1, 1)
plt.stem(linear_conv)
plt.title('Linear Convolution')
plt.xlabel('Sample')
plt.ylabel('Amplitude')

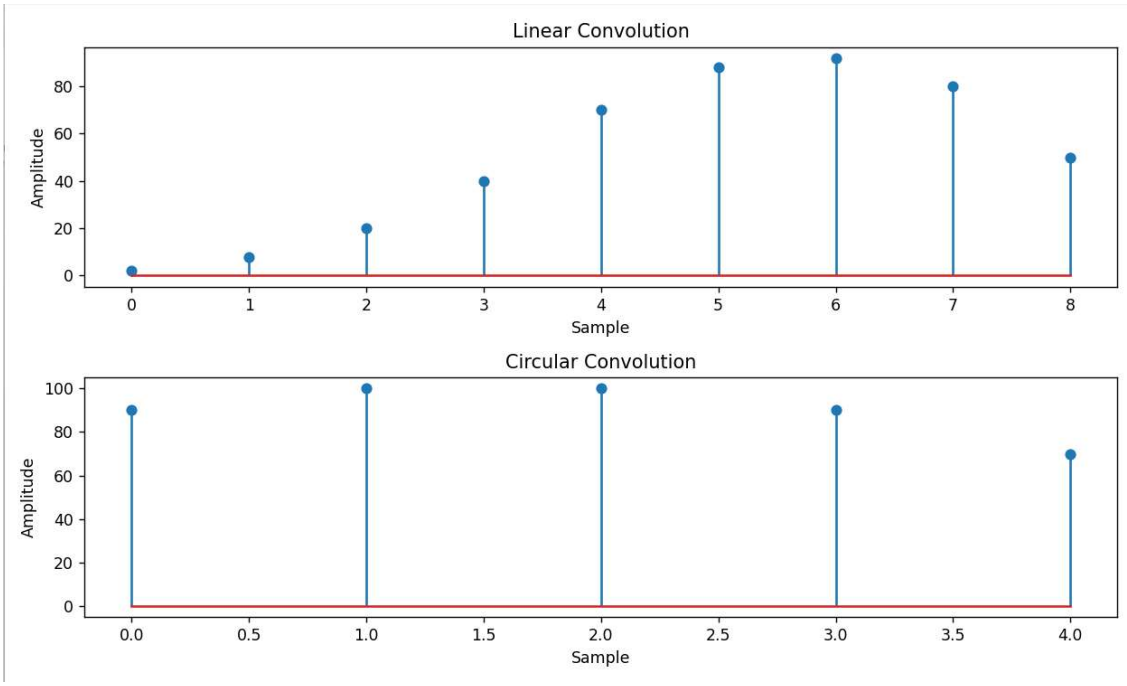
plt.subplot(2, 1, 2)
plt.stem(circular_conv)
plt.title('Circular Convolution')
plt.xlabel('Sample')
```

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```
plt.ylabel('Amplitude')

plt.tight_layout()
plt.show()
```

Output :-



Conclusion :- This experiment successfully demonstrated the simulation of **linear convolution** and **circular convolution** on discrete-time signals using Python. Linear convolution produced an output of longer length, while circular convolution generated an output with the same length as the input signals. The results were visualized using plots, making it easy to compare both convolution methods. This experiment helped in understanding the practical implementation and differences between linear and circular convolution in digital signal processing.